# Make Africa Machine-Readable

An AI model can explain the physics of a landslide. But it cannot tell a lithium mine manager whether a pit wall has shifted overnight, which fractures are widening, or whether workers can safely enter the excavation without current surveys, geotechnical measurements and a reliable history of the site.

**A model can understand the physics of a collapsing rock face without knowing that the rock face is collapsing.**

This is worth remembering amid the enthusiasm for African AI factories, cloud regions and locally hosted language models. Those investments address genuine constraints. But a model's ability to generate plausible answers is not the same as an accurate, current understanding of the physical environment in which it is expected to act.

The missing capability is an **observation layer**: systems that capture, verify and update information about the physical world. This problem exists everywhere, but becomes more consequential when records are fragmented and capital is scarce.

## A model has never inspected your transformer

Language models learn general patterns from text and code; vision systems learn to interpret imagery, while forecasting models extract relationships from historical observations. Increasingly, these capabilities overlap within multimodal AI systems. But knowledge learned during training is not the same as reliable information about a particular physical system operating today.

Fine-tuning can adapt a model to an industry, but it cannot establish the present condition of equipment it has never observed. Synthetic data can help simulate missing information, but predictions still need to be tested against physical reality. AI can infer what it cannot directly measure, provided there is enough related evidence to make those estimates credible.

This is where the challenge becomes interesting. To understand an electricity network, AI needs more than images of power lines. It needs observations that can be associated with identifiable assets: geospatial coordinates, network connections, sensor readings, maintenance histories and the outcomes of previous interventions. An image of a pole, a voltage reading and a fault report become far more useful when they can be reliably linked to the same physical asset.

The challenge, then, is not simply collecting more data or building more sophisticated models. It is constructing and continuously updating a trustworthy representation of the physical world upon which those models can reason and act.

This is the practical meaning behind the co-opted term, **embodied cognition**. Intelligence becomes useful in the physical economy when perception, a representation of the environment, a decision and feedback form a working system. Robots are one expression of this. So are industrial inspection and grid planning.

## What is missing is not another map

Africa is not an unmapped blank space. National agencies, utilities, researchers and commercial providers already produce valuable datasets. Google's Open Buildings project, for example, has identified roughly 1.8 billion building footprints across Africa and other Global South regions. That is an enormous advance in geographical visibility.

But a building outline is not an electricity customer. It cannot establish whether the building is a clinic or workshop, whether it has a connection, how reliable that connection is or whether the nearest transformer has spare capacity. Resolution, image age, licensing and ground-level verification also

matter: the imagery adequate for counting buildings may be inadequate for assessing a small rooftop solar installation. The International Energy Agency has noted that electrification planners often lack utility information identifying which buildings are actually connected. Night-time light observations are useful proxies, but can miss small off-grid systems or imply that a lit street means every nearby building has power. An Oxford-led study in Kenya, Uganda and Sierra Leone found that volunteers identified 93% of rural homes in a mapping exercise, albeit with many false positives. A much faster automated detector found only 36%. Scale, accuracy and verification involve different costs; a prediction is not automatically an asset record. UC Berkeley's GridWatch research used independent sensors in Accra and reported substantial undercounting of outages in the monitored areas relative to utility records. This is a local finding, but it shows why independent measurement matters.

For physical infrastructure, I find it helpful to distinguish six levels of knowledge: **existence, location, identity, connectivity, condition and behaviour**. Satellite imagery may establish the first two. Field surveys and engineering records can establish the next two. Drones and specialist instruments help assess condition. Meters and operating data capture behaviour. The difficult part is joining those observations into one trustworthy, time-stamped account of an asset.

The shortage is of decision-grade knowledge about the physical economy.

## An electricity network that can learn

Consider a distribution utility planning to reinforce a feeder serving a growing industrial district. It has an electrification target and a capital budget. What it may not have is a reconciled picture of the lines, poles and transformers already there, their actual connectivity, their condition and their performance under load.

A sensible starting point is a limited survey, not a national digital twin. Existing GIS and work-order records are brought together. Satellite imagery describes the surrounding settlement. Drones inspect accessible overhead assets and vegetation. Local technicians verify identities, nameplates and circuit connections. A small number of monitors record voltage and interruptions where operational coverage is weak. The resulting network model can inform which inspections, repairs or extensions deserve attention first.

No single sensor performs the whole task. A drone photograph cannot diagnose an internal transformer winding fault. A voltage monitor cannot identify a cracked cross-arm. And a machine-learning classifier cannot replace a load-flow study. The value lies in linking unlike observations to the same asset and the same decision.

The same applies to water: a pressure anomaly is only useful if pipe maps, valve records and repair histories locate the likely fault. This is not an undiscovered market. Companies such as nLine and SteamaCo already provide grid monitoring and metering; new systems should make such sources work together, not needlessly replace them.

The economics remain demanding. Better information might reduce inspections, prevent outages or improve network expansion, but only if operators can act on it. Businesses may capture more of the value of reliable power than the utility paying for the survey. A useful AI application needs a buyer and a measurable outcome, not just an impressive map.

Here the relationship between infrastructure and intelligence runs in both directions. Better observation improves the next maintenance or investment decision. Newly installed assets can be built with appropriate metering and documentation. Their operating records then improve later decisions. **The process of building infrastructure can build the data infrastructure at the same time.**

This becomes a genuine feedback loop only when a decision is acted upon and its consequences are recorded. Otherwise, it is merely an archive of expensive observations.

## Sovereignty over the view

The case for observation is not a licence to map everything and claim ownership of the result. A survey of power lines may also capture homes and movement. Water meters can reveal household behaviour. Geospatial records can improve public services while exposing communities to surveillance, land disputes or exclusion. Critical-infrastructure data can create security risks of its own.

That is why data sovereignty cannot be reduced to keeping servers inside national borders, or to having an African company's name on the invoice. It concerns who commissions the observations, who can verify or correct them, who may train models on them, and who captures the value. The African Union Data Policy Framework already provides an important basis for thinking about interoperability, rights and equitable participation.

The extractive version of this business is easy to imagine: surveyors and communities supply knowledge of their surroundings; a company turns it into a proprietary model; the people who made the data accurate lose both access and influence over its subsequent uses. Replacing a foreign data monopolist with a domestic one would not make that arrangement equitable.

Publicly financed basemaps should be reusable where disclosure is safe and lawful. Customers should have clear rights to export and correct their own asset records. Personal and security-sensitive information requires stricter controls; derived commercial intelligence requires negotiated rights rather than assumed ownership of everything a camera can see. The local people doing the verification should participate in the resulting economic capability, not simply supply cheap labels for someone else's models.

The objective is to develop the capacity to observe physical systems, improve them measurably and govern the knowledge that results.

***If there is a point to all of this, it is this: if artificial intelligence is to become embodied in Africa's economy, its cognition must be grounded in the physical and social realities of that economy, and remain accountable to the people whose lives it shapes.***

***That, ultimately, is what I mean by embodied cognition.***